

Static Analysis and Verification

SAV 2020/2021

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Lattices and Fixpoints

A Brief Introduction

Partial Orders

- ❖ A tuple $(A, \leq_A)$ is a **poset** (partially-ordered set) iff A is a **set** and $\leq_A \subseteq A \times A$ is a **partial order** (i.e., a reflexive, transitive, and antisymmetric binary relation) on A .
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- ❖ An example: For $(2^{\{a,b,c\}}, \subseteq)$, $\sqcup\{\emptyset, \{a\}, \{b\}\} = \{a, b\}$, and $\{a\} \sqcap \{b, c\} = \emptyset$.

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- ❖ Given a poset $(A, \leq_A)$, a set $B \subseteq A$ is a **chain** iff $\forall b, b' \in B. b \leq_A b' \vee b' \leq_A b$.
 - E.g., $\{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}\}$ is a chain wrt. $(2^{\{a,b,c\}}, \subseteq)$.

Functions on Lattices

❖ Let $(A, \leq_A)$ and $(B, \leq_B)$ be lattices.

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 - 0 is the **least fixpoint** of f and ∞ is the **greatest fixpoint** of f .

Knaster–Tarski Theorem

- ❖ **Knaster–Tarski Theorem.** Let $(A, \leq_A)$ be a complete lattice and let $f : A \longrightarrow A$ be a monotonic function. Then the set of fixpoints of f in $(A, \leq_A)$ is also a complete lattice.
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For the more curious:

❖ In more constructive terms, the least fixpoint of f is the stationary limit of $f^\alpha(\perp_A)$ for α ranging over the ordinals.

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- Every ordinal can be represented as the set of all smaller ordinals. There is the **zero ordinal**, **successor ordinals**, and **limit ordinals**. Natural numbers correspond to the so called finite ordinals (ordering types of finite sets), the set of natural numbers is the first infinite ordinal, and so on.

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❖ A dual result holds for the **greatest fixpoint**.

Kleene Fixpoint Theorem

- ❖ **Kleene Fixpoint Theorem.** Let $(A, \leq_A)$ be a complete lattice and $f : A \longrightarrow A$ a function.
- If f is $\sqcup$ -continuous, the least fixpoint of f is $\mu f = \sqcup \{f^i(\perp_A) \mid i \geq 0\}$.

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 - Moreover, a $\sqcup$ -continuous function is monotone, and hence one in fact computes the supremum of the ascending chain
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- If f is $\sqcap$ -continuous, the greatest fixpoint of f is $\nu f = \sqcap \{f^i(\top_A) \mid i \geq 0\}$.
 - Moreover, a $\sqcap$ -continuous function is monotone, and hence one in fact computes the infimum of the descending chain
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❖ **Theorem.** For finite complete lattices, every monotonic function is $\sqcap$ - and $\sqcup$ -continuous.

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❖ **Theorem.** For finite complete lattices, every monotonic function is $\sqcap$ - and $\sqcup$ -continuous.

❖ **Corollary.** On finite lattices, the Kleene fixpoint theorem is applicable, hence,

- to compute the least fixpoint, start with $\perp_A$ and iteratively apply f till
$$f^i(\perp_A) = f^{i+1}(\perp_A) = \mu f,$$
- to compute the greatest fixpoint, start with $\top_A$ and iteratively apply f till
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